

Herald



Herald — Continuation

Field	Value
Revision	9
Created	2026-05-20
Last modified	2026-05-27
Status	active
Status summary	2026-05-27 checkpoint (HEAD f833b0b, all 4 mirrors synced). MAJOR FINDING: the 2nd-bot qaherald-auto automation is structurally impossible — Telegram never relays bot-to-bot messages in groups (proven; full forensic record docs/research/telegram-bot-to-bot-constraint.md, commit 5267f14). Real-channel inbound automation requires MTProto (a real user account via gotd/td) — OPERATOR DECISION PENDING (needs my.telegram.org app_id+app_hash + 1-time phone login). This session: (1) pulled HelixConstitution to 3a085b9 (gained §11.4.85 stress+chaos / §11.4.86 / §11.4.87 endless-loop-autonomous / §11.4.88 background-push); (2) propagated §11.4.85/87/88 into Herald CLAUDE/AGENTS/QWEN + new HERALD_CONSTITUTION §108 (merge 4888081); (3) built qaherald-auto framework COMPLETE — qaherald lifecycle Cobra cmd + MessengerClient interface (Telegram impl) + 15 scenarios + orchestrator + report + preflight (10 gates) + shell adapter, all hermetic tests PASS (T1-T8); (4) fixed preflight G1 false-negative (real getChatMember membership proof, b45e45d); (5) Wave 7 STARTED — T1 channels.Channel interface (688819a) + T2 registry+init (22ba96f) + T3 per-channel inbox (0080339) + T4 generalized self-filter (f833b0b) → HRD-110/111/112/113 closed. Prior: v0.4.0 (Wave 6 closed-loop, HRD-100 real human round-trip msg 25→26) + v0.5.0 (Wave 6.5 lifecycle, premature S1+S2-only tag) both pushed. Anti-bluff gates green; pre-existing carry-over: Postgres-SASL e2e FAILs (E7-E12/E14-E16/E37-E42 — local PG SASL, unrelated).
Issues	HRD-008, HRD-015, HRD-018 (in progress), HRD-019..HRD-027, HRD-029..HRD-056, HRD-081, HRD-085..HRD-090, HRD-114..HRD-121 (Wave 7 T5-T12 pending)
Issues summary	see Issues.md. Wave 7 closed HRD-110/111/112/113 this session; HRD-114..121 are the remaining Wave 7 tasks T5-T12.
Fixed	HRD-001..HRD-017 (less open), HRD-028, HRD-080, HRD-092..HRD-101, HRD-110, HRD-111, HRD-112, HRD-113 see Fixed.md

Field	Value
Fixed summary	Resume at Wave 7 T6 (docs/superpowers/plans/2026-05-27-wave7-generic-messenger.md line 980; HEAD 42448a2). Wave 7 done: T1 channels.Channel interface (reply method SendReplyGeneric) + T2 registry/init + T3 per-channel inbox + T4 channels.IsSelfEcho/StampSender + T5 multi-channel pherald listen (HRD-110..114 closed; inbound.TgramReplier→Replier generic-sig rename done via a channelReplier shim — tgram's native int SendReply untouched, channels.Channel.SendReplyGeneric bridged; multi-channel fan-in via sync.WaitGroup, NOT errgroup, go.mod unchanged; HERALD_CHANNELS=tgram, slack parsed, default tgram). Remaining: T6 Slack adapter (Socket Mode — VENDORS slack-go as a new submodule: verify tests/test_constitution_inheritance.sh I6 gate still passes after the .gitmodules entry; confirm slack-go v0.16.0 field names inner.Files/socketmode.EventTypeEventsAPI/UploadFileV2Parameters before coding; T5 pre-staged perChannelConfig("slack") + blank-import comment in listen.go — HERALD_CHANNELS=tgram, slack currently errors ErrUnknownChannel until T6 registers slack, by design) · T7 qaherald Slack MessengerClient · T8 spec §11.0/§32.2/§43 · T9 e2e E81-E88 · T10 paired mutation gate · T11 §11.4.85 stress+chaos · T12 docs/Issues→Fixed/v0.6.0/4-mirror push. THREE GATED/DEBT ITEMS: (a) MTProto real-channel automation — OPERATOR must provide my.telegram.org app_id+app_hash (folds into a qaherald MessengerClient MTProto impl reusing the entire scenario engine); operator should also /revoke the leaked @pherald_qa_bot token. (b) Wave 6.5 mutation-gate M3 STALE — tests/test_wave6.5_mutation_meta.sh M3 regex targets opts.ReplyTo but Wave 6.5 T6 renamed to textOpts.ReplyTo in send.go → M3 errors "anchor not found", non-functional gate (§107 bluff surface, FIX needed). (c) docs/Issues.md + docs/Fixed.md PDF/HTML/DOCX siblings stale after HRD-110..113 migrations — regen bash scripts/export_docs.sh docs/Issues.md docs/Fixed.md at Wave 7 close. Then: 39 remaining §42/§43 HRDs (018..027, 029..056, 081, 085..090, 008, 015) for v1.0.0; docs-audit #147. §11.4.87 endless-loop: continue until Issues.md zero-active + CONTINUATION §3 empty + no subagent in flight. NO BLUFF. Push 4 mirrors every 2-3 commits.
Continuation	

Table of contents

- [§0. How to use this document](#)
- [§1. Snapshot](#)
- [§2. Last commit landed](#)
- [§3. Active work](#)
- [§4. Next concrete steps](#)
- [§5. Long-form pointers](#)

§0. How to use this document

Paste the following block into any CLI agent (Claude Code / OpenCode / Cursor / Aider / Gemini CLI) to resume Herald work exactly where it was left:

You are working on the Herald project at ~/Projects/Herald (also reachable as the Herald/ submodule of a consuming project). The Helix Universal Constitution lives at <ancestor>/constitution/ (parent-walk discovery). Read in this order: CLAUDE.md, AGENTS.md, README.md, docs/guides/HERALD_CONSTITUTION.md, docs/guides/CONSTITUTION_INHERITANCE.md, docs/specs/mvp/specification.V3.md. Then read docs/CONTINUATION.md (this file) for live state, docs/Issues.md for open work, docs/Status.md for current phase, docs/Fixed.md for closed history. Go workspace builds via go test ./commons/... ./commons_prefix/... ./commons_messaging/... ./commons_storage/.... Inheritance gate tests/test_constitution_inheritance.sh MUST exit 0 before any commit. Multi-mirror fan-out push to four hosts (GitHub + GitLab + GitFlic + GitVerse) is mandatory per Constitution §103.

§1. Snapshot

- **Active spec:** docs/specs/mvp/specification.V3.md Revision 4 (~4300 lines).
- **Archived specs:** V1 + V2 in docs/specs/mvp/archive/ (frozen).
- **Go modules:** commons, commons_prefix, commons_messaging, commons_storage, pherald (all compile + unit tests pass).
- **Container scaffold:** quickstart/{docker-compose.quickstart.yml, Dockerfile.pherald, otel-config.yaml, .env.example} for §26.5 quickstart.
- **CLI:** pherald version --json returns canonical build info; serve/send/doctor/migrate/subscriber/deadletter stubbed with HRD-NNN pointers.
- **Inheritance gate:** 12 PASS / 0 FAIL. Meta-test ✓.
- **Phase:** implementation r1.

§2. Last commit landed

This commit (V3 r4 + Go scaffold + tracking-doc refresh) closes HRD-009/HRD-009b/HRD-013/HRD-014 (with a Universal §11.4.19 atomic Issues.md → Fixed.md migration) and lands spec V3 §37–§41 (tracker events, workable-item announcement contract, message presentation + Herald Canonical Template, docs/tests completeness, REST API surface). Builds + tests:

```
$ go test ./commons/... ./commons_prefix/... ./commons_messaging/... ./commons_storage/...
ok      github.com/vasic-digital/herald/commons 1.135s
ok      github.com/vasic-digital/herald/commons_prefix 0.639s
ok      github.com/vasic-digital/herald/commons_messaging/channels/null 0.000s
ok      github.com/vasic-digital/herald/commons_messaging/dispatch/claude 0.000s
ok      github.com/vasic-digital/herald/commons_storage 1.630s
```

§3. Active work

ID	Status	What
HRD-008	in_progress	Operator-side Quickstart compose validation (Postgres + Redis + OTel + pherald container) — scaffold shipped this commit; live end-to-end run pending operator.
HRD-010	open	

ID	Status	What
		commons_storage live (pgx pool + golang-migrate driver + River queue + Redis ACL).
HRD-011	open	Telegram adapter live (telebot SDK + getUpdates long-poll + webhook secret_token).
HRD-012	open	Claude Code dispatcher live (claude --resume + parse <<<HERALD-REPLY>>>).
HRD-015	open	Inheritance gate I8 invariants for Go scaffold (go.work + commons/types.go + null adapter test passes).
HRD-016	open	REST API surface via Gin Gonic per spec §41.
HRD-017	open	Propagate Universal §11.4.6X spec-versioning + submodule-catalogue-first mandates into the parent constitution.

§4. Next concrete steps

1. **HRD-018 Catalogue-Check survey + commons_constitution scaffold** — first action of the §42 implementation rollout. **MUST start** with a vasic-digital + HelixDevelopment catalogue survey per the brand-new Universal §11.4.74. Record Catalogue-Check: reuse|extend|no-match on HRD-018 before any Go code lands. Then scaffold the Evaluator interface, 12 event-class emit helpers, constitution_state + constitution_bindings migrations, bundle-hash captureer, mode-ladder runtime config.

2. **HRD-008 quickstart validation** — On a fresh laptop with Podman or Docker:

```
git clone <Herald repo>
git submodule update --init
cd quickstart
podman build -t pherald/pherald:dev -f Dockerfile.pherald ../..
cp .env.example ../../.env && $EDITOR ../../.env
podman-compose -f docker-compose.quickstart.yml up -d
curl --retry 30 --retry-delay 2 http://localhost:24090/readyz
```

The current pherald serve returns "not implemented" — HRD-010/HRD-011/HRD-012 must land first to make the live curl POST /v1/events succeed end-to-end. Validation reveals which of the spec's assumptions (port ranges, Compose syntax, OTEL collector config, Postgres healthcheck) hold against real infrastructure.

3. **HRD-010** — wire commons_storage/storage.go's MigrationDriver to golang-migrate; bring pgx + River + Redis client up; add integration tests under // go:build integration.
4. **HRD-011** — replace commons_messaging/channels/tgram/tgram.go stub with a live implementation against gopkg.in/telebot.v3 or github.com/mymrac/telego; recorded HTTP fixtures under testdata/.
5. **HRD-012** — replace commons_messaging/dispatch/claude_code/claude_code.go's Dispatch stub with a real claude --resume invocation; capture session UUID; parse <<<HERALD-REPLY>>>.
6. **HRD-016** — scaffold pherald/internal/http/ with Gin routes per V3 §41; wire pherald serve to mount the Gin router on http_port.

7. **HRD-017** — propagate Universal §11.4.6X new mandates (spec-versioning + submodule-catalogue-first) into the constitution submodule.

§4a. Wave 6 live-test handoff (T10b — gating tag v0.4.0)

Operator-supplied credentials required — Wave 6's e2e invariants E63-E70 land as honest SKIPs until the live closed-loop runs against real Telegram + real Claude Code with real chat messages exchanged.

Step-by-step.

1. Export credentials in the shell that will drive the test:

```
export HERALD_TGRAM_BOT_TOKEN="<your-bot-token-from-  
BotFather>"  
export HERALD_TGRAM_CHAT_ID="<numeric-chat-id-where-bot-is-  
admin>"  
export HERALD_CLAUDE_BIN="$(command -v claude)" # path to  
the claude CLI binary  
export HERALD_PROJECT_NAME="Herald" # or any  
operator-chosen session name
```

2. Pick a `<run-id>` (timestamp + 4-char nonce):

```
RUN_ID="$(date -u +%Y-%m-%dT%H-%M-%S)-$(uuidgen | cut -c1-4)"  
QA_DIR="docs/qa/HRD-100-${RUN_ID}"  
mkdir -p "${QA_DIR}/attachments"
```

3. Type ONE message in the configured Telegram chat (script reads it via `getUpdates` within 60s).
4. Run the live closed-loop test:

```
bash tests/test_wave6_live_loop.sh
```

The script builds `pherald`, observes the original `message_id`, starts `pherald listen --bot-token "$HERALD_TGRAM_BOT_TOKEN" --chat-id "$HERALD_TGRAM_CHAT_ID" --qa-out-dir "$QA_DIR"` in the background, waits up to 45s for a reply with `reply_to_message_id == original`, and exits 0 on PASS.

5. Capture supplementary logs:

```
cp /tmp/pherald-w6.log "${QA_DIR}/pherald-listen.log"  
# claude stdout/stderr — paths depend on the journaling  
setup. Copy whatever  
# the test produced into ${QA_DIR}/claude-stdout.log +  
claude-stderr.log.
```

6. Author a brief `${QA_DIR}/README.md` (5 lines minimum) narrating: who ran it, when, what message was sent, what reply came back, any anomalies. This is NOT auto-generated — operator-written narrative is the §107.x evidence anchor.

```
7. git add ${QA_DIR}/ && git commit -m "Wave 6 step 10b: docs/
   qa/HRD-100-${RUN_ID}/ live closed-loop evidence" then proceed to
   T13b (v0.4.0 tag + 4-mirror push).
```

If the script **SKIPS** (any env var unset): the test prints `SKIP: <reason>` and exits 0. That is honest §11.4.3 hardware-absent SKIP-with-reason, not a PASS. Tag `v0.4.0` **MUST NOT** be created until at least one PASS run is committed to `docs/qa/HRD-100-<run-id>/`.

§4b. qaherald-auto resume runbook (next session — gating tag v0.5.1)

Status as of 2026-05-23 session end:

- HEAD `8ad4cb9` on main; all 4 mirrors in sync (`github.com:vasic-digital/Herald.git`, `gitlab.com:vasic-digital/herald.git`, `gitflic.ru:vasic-digital/herald.git`, `gitverse.ru:vasic-digital/Herald.git`).
- Tag `v0.4.0` shipped (Wave 6 closed-loop). Tag `v0.5.0` shipped (Wave 6.5 lifecycle implementations — premature: only S1+S2 live evidence + carries the `pre-53a7ad3` envelope-action-guidance bug). Tag `v0.5.1` is the target after qaherald-auto T6-T8 land + full 15-scenario evidence committed.
- Plan: `docs/superpowers/plans/2026-05-23-qaherald-auto.md` (commit `52ce679`).
- Done: T1 skeleton + T2 messenger + T3 scenarios + T4 orchestrator + T5 report (combined commit `8ad4cb9` for T3-T5).
- TODO: **T6 pre-flight** + **T7 hermetic tests** + **T8 shell-adapter** then live run + tag.

Resume prerequisites

The following are **already provisioned** in `~/ .zshrc` (export-on-shell-start):

```
HERALD_TGRAM_BOT_TOKEN
    # @atmosphere_worker_bot — pherald-bot (Telegram bot
    8823384001)
HERALD_TGRAM_CHAT_ID      # -4946584787 (ATMOSphere Development
    group)
HERALD_OPERATOR_IDS      # 2057253161 (the operator's user-id)
HERALD_QA_BOT_TOKEN      # @pherald_qa_bot — 2nd bot
    (8971749017, privacy DISABLED, member of the group)
```

The qa-bot token is plaintext in `~/ .zshrc` — operator should `/revoke` via `@BotFather` after Wave 6.5 close-out and regenerate (the token leaked into one prior session's bash output earlier today, was never committed to git but lives in `/private/tmp/claude-501/...` operator-local files).

The containers `herald-postgres` (port 24100) and `herald-redis` (port 24200) are LIVE — both have been Up since the start of this session; data persists across podman restarts. Schema is at migration v12 (all 12 applied).

Resume sequence (suggested next-session order)

1. Pull state + verify:

```
cd ~/Projects/Herald && git pull origin main && git log --
    oneline -10
git status --short # expect clean
git stash list      # 2 stashes preserved: T8 Wave-5 salvage
    + premature S3-S5 transcript WIP
```

2. **Spawn T6 (pre-flight)** via a fresh subagent — read docs/superpowers/plans/2026-05-23-qaherald-auto.md Task 6 in full. Implementation lands at qaherald/internal/lifecycle/preflight.go + preflight_test.go. 10 gates: pherald-bot reachable via getMe, qa-bot reachable, qa-bot privacy disabled, qa-bot is group member, OPERATOR_IDS contains qa-bot, etc. Each gate has distinct exit code for diagnostics.
3. **Spawn T7 (full hermetic tests)** — extends T3-T5's 12 tests with httptest-based scenario simulation. Coverage targets: every scenario PASS path + every FAIL diagnostic + S9 SKIP path + S14 outbound-attachment sha256 round-trip + S11/S12/S13 inbound-attachment download + Issues.md/Fixed.md fs-mutation assertion.
4. **Spawn T8 (shell-script adapter)** — rewrites tests/test_wave6.5_lifecycle.sh to delegate to qaherald lifecycle for automated runs. Keeps the original --manual flag for the operator-typing interactive UX as a fallback.
5. **Run the live e2e (the §107 watershed):**

```
cd ~/Projects/Herald
# Add qa-bot user-id to OPERATOR_IDS so S5/S6/S8/S10 succeed
export HERALD_OPERATOR_IDS="$
    {HERALD_OPERATOR_IDS},8971749017"
# Re-export from .zshrc (or just open a new shell)
source ~/.zshrc

# Build pherald + qaherald
go build -o /tmp/pherald ./pherald/cmd/pherald
go build -o /tmp/qaherald ./qaherald/cmd/qaherald

# Start pherald listen in background with QA journaling
RUN_ID="$(date -u +%Y-%m-%dT%H-%M-%S)-w6.5live"
PHERALD_QA_DIR="docs/qa/HRD-101-lifecycle-`${RUN_ID}`-pherald"
QAUTO_QA_DIR="docs/qa/HRD-101-lifecycle-`${RUN_ID}`"
mkdir -p "${PHERALD_QA_DIR}/attachments" "${QAUTO_QA_DIR}/
    attachments"
/tmp/pherald listen --qa-out-dir "${PHERALD_QA_DIR}" --docs-
    dir docs &
PHERALD_PID=$!
trap 'kill -TERM $PHERALD_PID 2>/dev/null' EXIT
sleep 5

# Run qaherald lifecycle (T6-T8 must be done first)
/tmp/qaherald lifecycle \
    --pherald-bot-username=atmosphere_worker_bot \
```

```
--pherald-qa-out-dir="${PHERALD_QA_DIR}" \
--out="${QAUTO_QA_DIR}" \
--run-id="${RUN_ID}"
# Exit 0 on all-PASS or all-PASS-with-S9-SKIP. Non-zero on
any FAIL.
```

6. Capture evidence:

- `${PHERALD_QA_DIR}/transcript.jsonl` — pherald's view (classifications, dispatch, replies)
- `${QAUTO_QA_DIR}/transcript.jsonl` — qaherald-auto's view (sends, assertions)
- `${QAUTO_QA_DIR}/report.md` — Markdown summary
- `${QAUTO_QA_DIR}/attachments/<sha256>.<ext>` — content-addressed inbound + outbound attachments
- `docs/Issues.md` + `docs/Fixed.md` — mutated by S5/S6/S8/S10 (and reverted by S10 + S15 cleanups; both files end in valid state)

7. Author operator README + commit:

```
cat > "${QAUTO_QA_DIR}/README.md" <<EOF
# HRD-101 Wave 6.5 lifecycle evidence — <run-id>

<5+ lines narrating: who ran it, when, which scenarios
      PASSed/FAILed/SKIPped,
bot reply quality observations, any anomalies. Operator-
      written, NOT auto-generated.>
EOF
git add docs/qa/HRD-101-lifecycle-${RUN_ID}*
git commit -m "Wave 6.5 step 13: HRD-101 live 15-scenario
      lifecycle evidence (qaherald-auto)"
```

8. Tag v0.5.1 + 4-mirror push:

```
git tag -a v0.5.1 -m "Wave 6.5 close: full lifecycle live
      evidence + envelope action-guidance fix"
git push origin main
git push origin v0.5.1
# Verify all 4 mirrors converged on the same SHA + tag
for mirror in github.com:vasic-digital/Herald.git
      gitlab.com:vasic-digital/herald.git
      gitflic.ru:vasic-digital/herald.git
      gitverse.ru:vasic-digital/Herald.git; do
  git ls-remote "git@${mirror}" v0.5.1
done
```

Open / known constraints for tomorrow

- **S9 (non-operator Done: rejection):** requires `HERALD_QA_BOT_TOKEN_NON_OPERATOR` env (a THIRD bot account whose user-id is NOT in `OPERATOR_IDS`). Without it, S9 emits a `SKIP-with-reason`. Operator may

register a third bot OR accept the SKIP (S9's logic is already unit-tested hermetically in T5's mutation gate).

- **First-scenario CC bootstrap:** ~30s on first inbound message because the Claude Code session must spawn. Subsequent scenarios use the cached session (~5-15s typical).
- **Issues.md mutation race:** if you run `qa Herald lifecycle` while ANOTHER pherald listen is also processing in the same checkout, Issues.md may receive double-writes. Single-pherald-listen invocation is the safe default.
- **Pre-existing carry-over:** 15 `e2e_bluff_hunt` FAILs on Postgres-SASL invariants (E7-E12 + E14-E16 + E37-E42). Root cause is local-PG container's SASL handshake; not related to Wave 6/6.5; can be addressed in a separate HRD post-v1.0.0.

Safe-shutdown checklist (NOW — end of 2026-05-23 session)

Check	State
All committed work pushed to 4 mirrors	YES — HEAD 8ad4cb9 confirmed on github+gitlab+gitflic+gitverse
Local working tree clean	YES — <code>git status --short</code> empty
Stashes preserved	2 stashes: T8 Wave-5 salvage (older) + premature mid-test S3-S5 transcript WIP (newer, will be regenerated cleanly by qa Herald-auto in next session)
Background pherald/qa Herald processes	NONE running (<code>ps aux grep pherald</code> empty)
Container state	<code>herald-postgres + herald-redis</code> Up 17 hours — persist data across <code>podman stop && podman start</code> ; can be left running OR stopped (<code>podman stop herald-postgres herald-redis</code>); next session restart is <code>podman start herald-postgres herald-redis</code>
/tmp/*.log files	one operator-token leak shred-deleted earlier today; remaining transient logs contain no credentials (verified)
.zshrc	contains 4 <code>HERALD_*</code> exports including <code>HERALD_QA_BOT_TOKEN</code> plaintext — operator should /revoke + regenerate via @BotFather post-Wave-6.5 closure
Memory entries	All session findings persisted under <code>/Users/milosvasic/.claude/projects/-Users-milosvasic-Projects-Herald/memory/</code> — survive across sessions

Machine is safe to shut down. Resume tomorrow by `cd ~/Projects/Herald && git pull origin main` and start with `qa Herald-auto` T6.

§5. Long-form pointers

- `docs/specs/mvp/specification.V3.md` — full active spec (Revision 4).
- `docs/specs/mvp/specification.V3.md#30-v2-self-review-log` — every review pass.
- `docs/guides/HERALD_CONSTITUTION.md` — §101..§106 extending Universal.
- `docs/guides/CONSTITUTION_INHERITANCE.md` — parent-discovery + gate.
- `tests/test_constitution_inheritance.sh` — the gate.
- `quickstart/` — HRD-008 scaffold.
- `commons/types.go` — the §11.0 type contract reference.